Lab Report: Support Vector Machines for Fashion-MNIST Classification

1. Binary SVM with Non-linear Kernels

To handle the non-linear nature of image data, this implementation utilizes the Gaussian Radial Basis Function (RBF) Kernel. This kernel projects the 784-dimensional input space into an infinite-dimensional feature space, allowing for the construction of a hyperplane that separates classes which are not linearly separable in the original pixel space.

The training process involves solving the dual Lagrangian optimization problem using the cvxopt library. The objective is to find the Lagrange multipliers a that minimize:

$\min_{\mathbf{a}} \frac{1}{2}\mathbf{a}^{\mathrm{T}}(\mathbf{TKT})\mathbf{a} - \mathbf{1}^{\mathrm{T}}\mathbf{a}$

Subject to:

$0 \leq a_i \leq C$

$\mathbf{a}^{\mathrm{T}}\mathbf{t} = 0$

Where $\mathbf{T}$ is a diagonal matrix of labels and $\mathbf{K}$ is the kernel matrix.

2. Predictive Model Implementation

The predictive model relies on the support vectors identified during training (where $a_i > 0$). For a new input $\mathbf{x}$, the decision function is:

$y(\mathbf{x}) = \operatorname{sgn}\left(\sum_{n \in S} a_n t_n k(\mathbf{x}, \mathbf{x}_n) + b\right)$

The bias term b is calculated using support vectors that lie strictly on the margin ($0 < a_n < C$). For these points, the prediction should exactly equal the label t_n. We average the bias across all such points for numerical stability:

$b = \frac{1}{N_S} \sum_{n \in S} \left(t_n - \sum_{m \in S} a_m t_m k(\mathbf{x}_n, \mathbf{x}_m)\right)$

3. Comparison of Voting Schemes

Because SVMs are inherently binary, two multiclass strategies were implemented:

Scheme	Mechanism	Result Summary
One-vs-Rest (OvR)	Trains 10 classifiers (one per class). Picks the class with the highest decision score.	Faster training but can result in tie regions where multiple classifiers claim a point.
One-vs-One (OvO)	Trains 45 classifiers (every pair). Class with the most votes wins.	More computationally expensive to train but typically more accurate for similar clothing items.

Experimental results showed that One-vs-One provided better separation for visually similar categories like Shirt vs. T-shirt, as the decision boundary only had to focus on the nuances between those two specific classes.

4. Hyperparameter Tuning Strategy

The model's performance is governed by two parameters: C and γ.

Gamma (γ): We used the heuristic $\gamma = 1/\text{n_features}$. This controls the width of the Gaussian; a value too high leads to overfitting (high variance), while too low leads to underfitting (high bias).

Penalty (C): We selected a high C (10.0). This choice prioritizes minimizing training errors over maximizing the margin width. Given the high dimensionality of the Fashion-MNIST data, a stricter penalty was necessary to ensure the model captured the specific structural details of the clothing items.

5. Multiclass Confusion Matrices

The confusion matrices highlight specific areas of difficulty.

- High Accuracy: Trouser, Bag, and Ankle Boot consistently show high recovery rates (>94%) due to their unique silhouettes.
- Confusion Clusters: Significant misclassification occurs between Shirt, T-shirt, and Pull-over. These items share similar pixel distributions around the neckline and sleeves, making them the primary source of error in the 96% recovery goal.